



Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

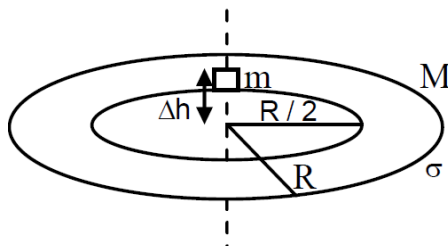
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A small block of mass m lies above a thin disk of total mass M , constant surface density σ , and radius R . A hole of radius $R/2$ is cut into the center of the disk, allowing the mass to pass through. The mass starts at a position along the central axis of the disk but is displaced a height Δh from the plane of the disk, where $\Delta h \ll R$. There are no other external forces at work on this system; i.e., the disk and block are out in deep space. Assume $m \ll M$. For the first order in Δh the block undergoes simple harmonic motion with respect to the centre of the plane of the disk, compute the period of oscillation of block.



(A) $T = \sqrt{\frac{2\pi R}{G\sigma}}$

(B) $T = \sqrt{\frac{\pi R}{G\sigma}}$

(C) $T = \sqrt{\frac{\pi R}{2G\sigma}}$

(D) None of these

2. Consider a metal can placed coaxially inside a long solenoid. The metal can be modelled as a thin cylindrical shell of radius r . The magnetic field due to the solenoid varies with time t as kt where k is a positive constant. The resistivity of the can is ρ and its breaking stress is σ . Ignore the self-inductance of the can and any currents induced in the flat surfaces of the can. Assume that the magnetic field produced by the induced currents is negligible. Compute the time at which the can breaks.

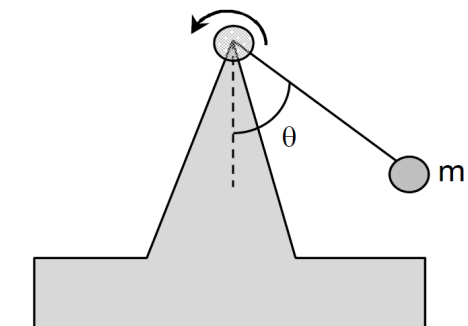
(A) $t = \frac{\sigma \rho}{r^2 K^2}$

(B) $t = \frac{2\sigma \rho}{r^2 K^2}$

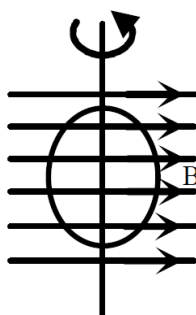
(C) $t = \frac{\sigma \rho}{2r^2 K^2}$

(D) $t = \frac{\sigma \rho}{3r^2 K^2}$

3. A motor is installed at the top of a pole rigidly fixed on a platform. A light rod of length $r = 1$ m is rigidly attached to the motor shaft at its one end and at the other end a small ball of mass m is attached. The rod can rotate in a vertical plane with the help of the motor. Total mass of the platform, the pole and the motor is $\eta = 4$ times the mass of the ball. The motor rotates the rod at a constant angular velocity. The platform is placed on a horizontal surface, where coefficient of friction is $\mu = 1/\sqrt{3}$. At which minimum angular velocity ω_0 of the rod, will the platform start sliding?



- (A) $\omega_0 = 2 \text{ rad/s}$
 (B) $\omega_0 = 3 \text{ rad/s}$
 (C) $\omega_0 = 5 \text{ rad/s}$
 (D) None of these
4. A ring of mean radius r and cross-sectional area A is made of a perfectly conducting wire. Inductance L of the ring is so small that inertia of free electrons cannot be neglected in the current building process. The free electron density in the conductor is n , mass of an electron is m and modulus of charge on an electron is e . Initially the ring is placed in a uniform magnetic field with its plane parallel to induction vector $\vec{B}$ of the field as shown in the figure. Find the current ring after it is turned through an angle 90° .



- (A) $i = \frac{B\pi r^2}{L + \frac{m \times \pi r}{ne^2 A}}$
 (B) $i = \frac{B\pi r}{L + \frac{m \times \pi r}{ne^2 A}}$
 (C) $i = \frac{B\pi r^2}{L + \frac{m \times 2\pi r}{ne^2 A}}$
 (D) $i = \frac{B \times 2\pi r}{L + \frac{m \times \pi r^2}{ne^2 A}}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

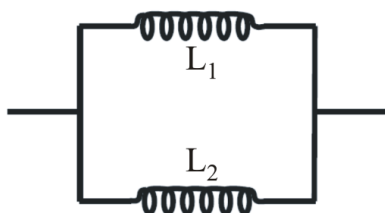
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

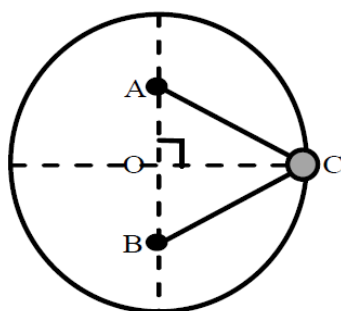
5.



For the given figure consider their mutual inductance is M then

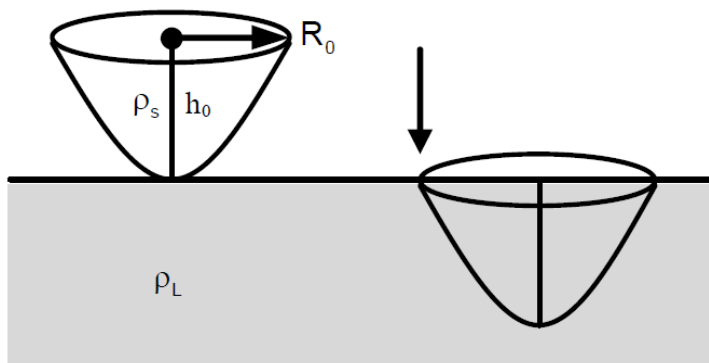
- (A) $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$ When mutual inductance is not opposing
- (B) $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is not opposing.
- (C) $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is opposing.
- (D) $L_{eq} = \frac{L_1 L_2 - M^2}{2L_1 + L_2 + 2M}$ When mutual inductance is opposing.

6. In a photoelectric effect experiment, a point source of light of power 40W emits mono-energetic photons of wavelength λ_1 that can just emit photoelectrons from an isolated metallic sphere of radius 1 cm, placed at a distance of 1 km from the light source. Now, three other sources of wavelength λ_2, λ_3 and λ_4 which are emitting the same number of photons as that of λ_1 brought near the source of λ_1 . Assume photo efficiency of 10^{-6} (Take $\lambda_1 = 4960\text{\AA}$, $\lambda_2 = 4133.33\text{\AA}$, $\lambda_3 = 5000\text{\AA}$, $\lambda_4 = 7200\text{\AA}$ and $hc = 12.400 \text{ eV}\cdot\text{\AA}$)
- (A) Number of photoelectrons emitted from the sphere per second is 2.5×10^9
- (B) The potential of the sphere when emission of photoelectron stops is 0.8 V.
- (C) The potential of the sphere when emission of photoelectrons stops is 0.5 V.
- (D) The time after which the emission of photoelectrons will stop is $1.39 \times 10^{-3} \text{ sec}$.
7. A dog running with a constant speed v is chasing a cat that is running with a constant velocity $\vec{u}$. During the chase, the dog always heads towards the cat. At an instant, direction of motion of the dog makes angle θ with that of the cat and the distance between them is r .
- (A) The magnitude of acceleration of dog at this instant is $\frac{uv \sin \theta}{r}$.
- (B) Radius of curvature of path followed by dog at this instant is $\frac{vr}{u \sin \theta}$.
- (C) The magnitude of acceleration of dog at this instant is $\frac{uv \cos \theta}{r}$.
- (D) Radius of curvature of path followed by dog at this instant $\frac{ur}{v \cos \theta}$.
8. A small bead of mass m can slide on a frictionless fixed ring of radius r . With the help of two identical strings of force constant k , the bead is connected to two nails A and B each on a diameter at a distance $0.5 r$ from centre O of the ring. Relaxed length of each string is negligible as compared to the radius of the ring. The bead is given a small velocity at point C. What can you predict about subsequent motion of the bead before any of the string strikes a nail? [ignore gravity]

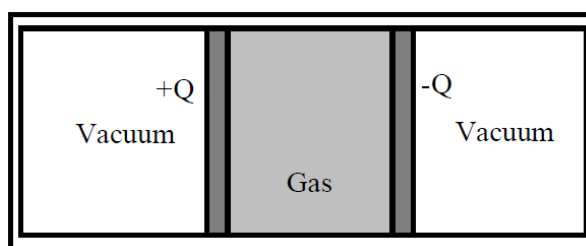


- (A) It will keep moving with its initial speed.
- (B) It will oscillate simple harmonically about the point C.
- (C) Its angular momentum about centre O of the ring is conserved.
- (D) Total elastic potential energy stored in both the strings is $1.25kr^2$.

9. A solid paraboloid of base radius R_0 and height h_0 is having uniform volume mass density of ρ_s is Inverted and placed just above the surface as shown, of a liquid having volume mass density of ρ_L . When paraboloid is released from rest it has been observed that when it becomes completely submerged in the liquid the paraboloid comes to the rest again. There exist uniform gravity of g . Assume that the liquid body is very large so that liquid level is not changing as the paraboloid enters into the liquid.



- (A) The phenomenon is true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.
- (B) The phenomenon is not true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.
- (C) The ratio of $\frac{\rho_L}{\rho_s}$ is 2.
- (D) The ratio of $\frac{\rho_L}{\rho_s}$ is 3.
10. A ideal gas having f degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of $+Q$ and $-Q$. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased k times with the help of an external agent. Then which of the following statements are correct



- (A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2 + f}{2 + f}$.
- (B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2 + f}{2 + f}$.
- (C) Work done by the agent in the process is $2(K^2 - 1)nRT_1$ where T_1 is initial temperature.
- (D) Work done by the agent in the process is $(K^2 - 1) \frac{Q^2 L_1}{A \epsilon_0}$ where L_1 is initial separation between pistons.

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

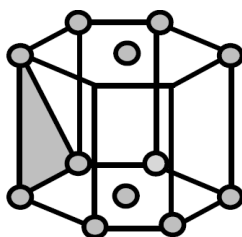
Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

There is a hollow prism with hexagonal base of side L_0 and height H_0 as shown. There are identical non conducting uniformly charged solid spheres are placed at each corner and two identical spheres at the centre of flat hexagonal faces. The charges spheres have uniform density of d_0 and radius of R_0 . What will be the flux passing through the shaded region and average charge density (ϕ) hexagonal prism. It is given that the flux passing through the hexagonal flat surface is k times the total flux passing through the prism.

(Take $\frac{R_0^3 d_0}{\epsilon_0} = 1$ SI units & $\frac{d_0 R_0^3}{L_0^2 H_0} = 1$ SI units.)



1. The value of ϕ_s is _____
2. The value of d is _____

Paragraph for Questions 3 and 4

One side of a thin metal plate is illuminated by the sun. When the air temperature is T_0 , the temperature of the illuminated side is T_1 and that of the opposite side is T_2 . The temperature of each sides be T_1' & T_2' , if another plate of double thickness is used. (Take $T_0 = 300$ K, $T_1 = 330$ K, $T_2 = 320$ K)

3. The value of T_1' (K) is _____
4. The value of T_2' (K) is _____

Paragraph for Questions 5 and 6

A boat is travelling in a river with a speed of 10 m/s along the stream flowing with a speed of 2 m/s. From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45 mm. Assume that attenuation of sound in water and air is negligible. The frequency detected by a receiver kept inside the river downstream is f_0 and the transmitter and the receiver are now pulled up into air. The air is blowing with a speed 5 m/sec in the direction opposite the river stream the frequency of the sound detected by the receiver is f_1 . (Temperature of the air and water = 20°C; Density of river water = 10^3 kg/m^3 ; Bulk modulus of the water = $2.088 \times 10^9 \text{ Pa}$; Gas constant $R = 8.31 \text{ J/mol} \cdot \text{K}$; Mean molecular mass of air = $28.8 \times 10^{-3} \text{ kg/mol}$; C_p/C_v for air = 1.4)

Note: Boat velocity is with respect to ground & receiver is stationary w.r.t ground

5. The value of f_0 (KHZ) is _____
6. The value of f_1 (KHZ) is _____

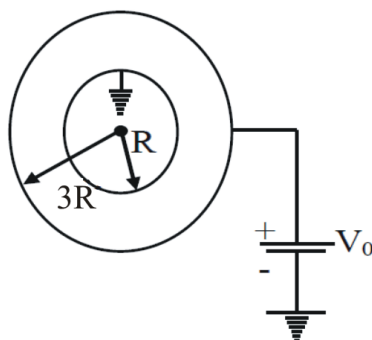
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

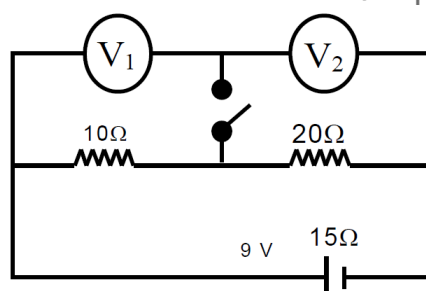
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

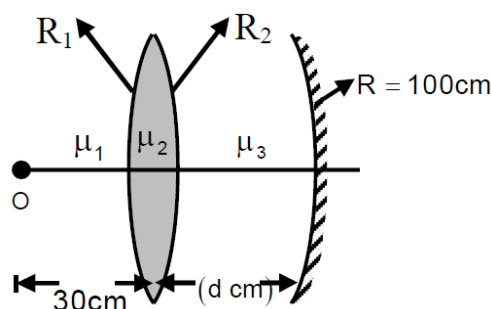
7. A conducting sphere with inner radius R and outer radius $3R$ has some alloys mixed in it due to which its resistivity changes with radial distance r according to the function $\rho_r = Kr$, where K is a +ve constant. Inner surface of the sphere is grounded while outer surface of sphere is maintained at potential V_0 with the help of a battery of emf V_0 . Battery is ideal. The potential of the sphere at radial distance of $2R$ is $V_0 \frac{\ln a}{\ln b}$. Find $(a + b)$.



8. In the circuit shown in the figure the electromotive force of the battery is 9 V and its internal resistance is 15Ω . The two identical voltmeters can be considered ideal. Let V_1 and V'_1 reading of 1st voltmeter when switch is open and closed respectively. Similarly, V_2 and V'_2 be the reading of 2nd voltmeter when switch is open and closed respectively. Then find the value of $\frac{V'_2 - V_2}{V_1 - V'_1}$.



9. The given thin lenses have $R_1 = R_2 = 10$ cm and $\mu_1 = 1$, $\mu_2 = 2$ and $\mu_3 = 4$. If final image formed at object and value of 'd' is $x \times 10$ cm. Find 'x'.



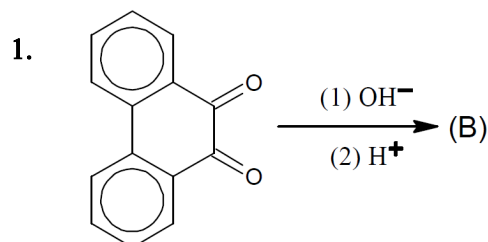
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

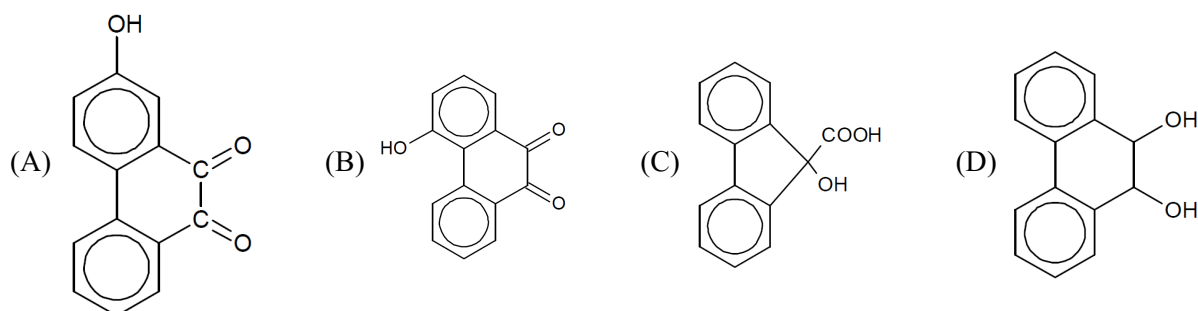
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases



The compound (B) would be



2. How many methoxy group present in the molecule of a compound (A) would be present by the result of a Zeisel method. Treatment of compound (A) $C_{20}H_{11}O_4N$ with hot conc. HI gives CH_3I indicating presence of methoxy group. When 4.24 mg of a compound (A) is treated with hot conc. HI and CH_3I thus produced is digested with alc. $AgNO_3$, 11.62 mg of AgI is obtained.

- (A) 1
(B) 2
(C) 3
(D) 4

3. The blue colour produced on adding H_2O_2 to acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution and shaking with ether is due to the formation of
- (A) Cr_2O_3 (B) CrO_4^{-2}
(C) CrO_3 (D) CrO_5
4. A 0.001 molal solution of a complex $[\text{MA}_8]$ in water has the freezing point -0.0054°C . Assuming 100% ionization of the complex salt and K_f for $\text{H}_2\text{O} = 1.86 \text{ K-Kg mol}^{-1}$, write the correct representation of the complex.
- (A) $[\text{MA}_8]$
(B) $[\text{MA}_7]\text{A}$
(C) $[\text{MA}_6]\text{A}_2$
(D) $[\text{MA}_5]\text{A}_3$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Cation(s) which form an ammine complex ion with excess of NH_3 is/are

(A) Ag^+

(B) Cu^{+2}

(C) Co^{+2}

(D) Al^{+3}

6. Which of the following statements regarding P_4O_{10} is correct ?

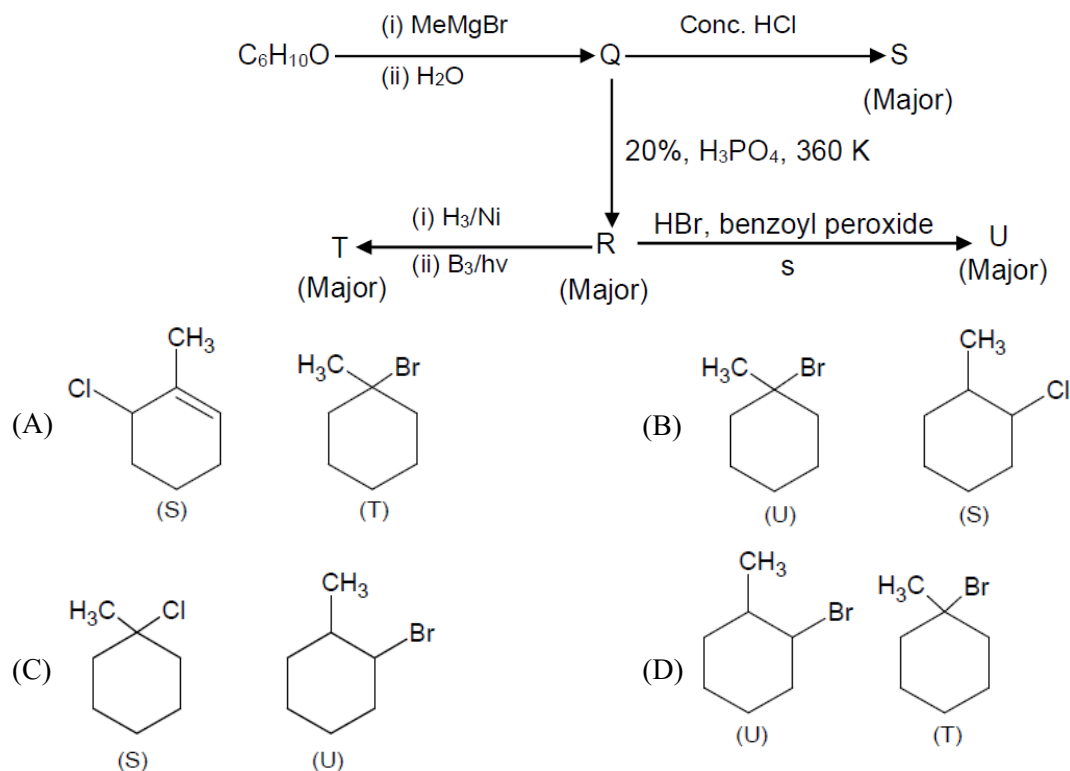
(A) It is an anhydride of metaphosphoric acid

(B) It is a strong dehydrating agent

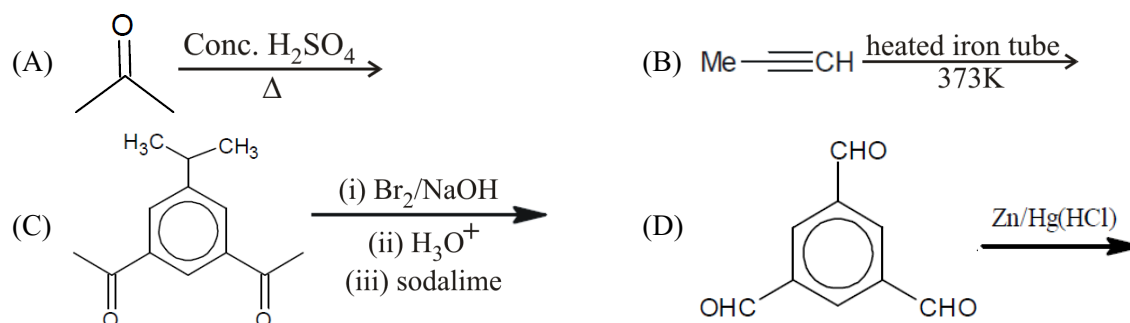
(C) In P_2O_5 , each phosphorous is attached to three oxygen atoms through single bond and also with another oxygen atom through double bond.

(D) The dehydrated product of HNO_3 obtained by treating it with P_4O_{10} is NO_2 .

7. Choose the correct option(s) for the following set of reactions



8. The reaction(s) leading to the formation of 1, 3, 5 trimethyl benzene is/are



9. Which of the following statement(s) is/are correct?

- (A) If temperature coefficient is greater than zero, cell reaction is endothermic.
 (B) If temperature coefficient is less than zero, cell reaction is endothermic.
 (C) If temperature coefficient is less than zero, cell reaction is exothermic.
 (D) If E_{cell} is negative then ΔG is negative and cell reaction is spontaneous.

10. In which of the following pair of solution will the values of Vant Hoff factor be same?

(Assuming same dissociation)

- (A) 0.05 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ and 0.10 M FeSO_4
 (B) 0.10 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ and 0.05 M $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (C) 0.20 M NaCl and 0.10 BaCl_2
 (D) 0.05 M $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ and 0.02 M $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

SECTION-II (i) : (Maximum Marks: 12)

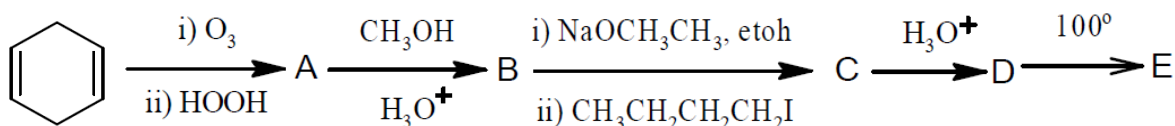
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

For the following reaction scheme



1. The molecular weight of B is
2. The molecular weight of E is

Paragraph for Questions 3 and 4

A 10^{-3} molal aqueous solution of a chromium complex having same number of ammonia molecule and Cl ions freezes at -0.00558°C . If all the ammonia molecules are acting as ligands and the percent of hydrogen in the complex is 4.6 (Given $K_f(\text{H}_2\text{O}) = 1.86\text{K Kg mol}^{-1}$) and assume 100% ionization self.

3. Find the number of Cl atom in complex is ____.
4. The value of Vant Hoff factor is ____.

Paragraph for Questions 5 and 6

The isoelectric point (P_I) is the P_H at which the amino acid exists only as a dipolar ion with net charge zero.

Structure of lysine and aspartic acid are $\text{C} \overset{\ominus}{\text{O}} \text{O} \text{CH} \left(\overset{+}{\text{N}} \text{H}_3 \right) (\text{CH}_2)_4 \text{NH}_2$ and

$\text{COO}^- \text{CH} \left(\overset{+}{\text{N}} \text{H}_3 \right) \text{CH}_2 \text{COOH}$ respectively. The $P_{K_{a1}}$, $P_{K_{a2}}$ and $P_{K_{a3}}$ of the dictation lysine are 2.18, 8.95 and 10.53 respectively. The $P_{K_{a1}}$, $P_{K_{a2}}$ and $P_{K_{a3}}$ of the cation of aspartic acid are 1.88, 3.65 and 9.60 respectively.

5. Isoelectric point lysine is _____.
6. Isoelectric point of Aspartic acid is _____.

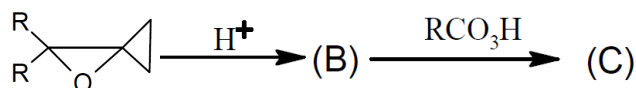
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

- Find the quantum numbers of the excited state of electrons (Let n_1 and n_2) in He^+ ion which on transition to ground state emit two photons of wavelength 30.4 nm and 108.5 nm respectively. Find the sum of $(n_1 + n_2) = ?$
- A compound AB forms NaCl type crystals and another compound XY forms CsCl type crystals. The formula. Mass of XY is three times that of AB but the cubic edge length of unit cell of AB crystals is twice that of XY. Calculate the ratio of density of XY to that of AB.
- Size of ring in product formed C is ?



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. Polynomial function $f(x)$ satisfying the condition $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If $f(10) = 1001$, then $f(20)$ is
- (A) 7001 (B) 8001
(C) 8000 (D) None of these
2. Let $f(x) = e^{\{e^{|x|} \operatorname{sgn} x\}}$ and $g(x) = e^{\{e^{|x|} \operatorname{sgn} x\}}$, $x \in \mathbb{R}$ where $\{ \}$ and $[]$ denotes the fractional and integral part functions, respectively. Also $h(x) = \log(f(x)) + \log(g(x))$ then for real x , $h(x)$ is
- (A) An odd function
(B) An even function
(C) Neither odd nor an even function
(D) Both odd as well as even function
3. The length and foot of the perpendicular from the point $(2, -1, 5)$ to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$ are
- (A) $\sqrt{14}, (1, 2, -3)$
(B) $\sqrt{14}, (1, -2, 3)$
(C) $\sqrt{14}, (1, 2, 3)$
(D) None of these
4. If $x = 9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangent is
- (A) $9x^2 - 8y^2 + 18x - 9 = 0$
(B) $9x^2 - 8y^2 - 18x + 9 = 0$
(C) $9x^2 - 8y^2 - 18x - 9 = 0$
(D) $9x^2 - 8y^2 + 18x + 9 = 0$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which of the following options are correct ?

(A) $\left[({}^nC_0 + {}^nC_3 + {}^nC_6 + \dots) - \frac{1}{2}({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + \dots) \right]^2 + \frac{3}{4}({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + \dots)^2 = 1$

(B) If a and b are two positive numbers such that $a^5 b^2 = 4$ then the maximum value of $\log_{2^{1/5}}(a^2) \cdot \log_{2^{1/2}}(b^2)$ is equal to 4

(C) Constant term in $(((((x-2)^2 - 2)^2 - 2)^2 - 2) \dots 2)^2$ is equal to 2

(D) The coefficient of x^{24} in $\left(\frac{{}^{25}C_1}{{}^{25}C_0} - x \right) \left(x - 2^2 \frac{{}^{25}C_2}{{}^{25}C_1} \right) \left(x - 3^2 \frac{{}^{25}C_3}{{}^{25}C_2} \right) \times \left(x - 4^2 \frac{{}^{25}C_4}{{}^{25}C_3} \right) \dots$
 $\left(x - 25^2 \frac{{}^{25}C_{25}}{{}^{25}C_{24}} \right)$ is equal to 2925

6. $a_1, a_1, a_2, \dots$ are distinct terms of an A.P. We call (p, q, r) an increasing triad if a_p, a_q, a_r are in G.P. where $p, q, r \in \mathbb{N}$ such that $p < q < r$. If $(5, 9, 16)$ is an increasing triad, then which of the following options is/are correct
- (A) If a_1 is a multiple of 4 then every term of the A.P. is an integer
- (B) $(85, 149, 261)$ is an increasing triad
- (C) If the common difference of the A.P. is $\frac{1}{4}$, then its first term is $\frac{1}{3}$
- (D) ratio of $(4k + 1)$ th term and $4k^{\text{th}}$ term can be 4
7. If $\int \frac{e^{x-1}}{(x^2 - 5x + 4)} 2x dx = AF(x - 1) + BF(x - 4) + C$ and $F(x) = \int \frac{e^x}{x} dx$, then
- (A) $A = -2/3$
- (B) $B = (4/3)e^3$
- (C) $A = 2/3$
- (D) $B = (8/3)e^3$
8. Let OA, OB, OC be coterminous edges of a cuboid. If l, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them respectively, then the value of $\frac{\left(\frac{1}{l^2} + \frac{1}{m^2} + \frac{1}{n^2}\right)}{\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)}$ is less than
- (A) 2
- (B) 3
- (C) 4
- (D) 5

9. Let the equation of a straight line L in complex form be $\bar{a}z + a\bar{z} + b = 0$, where a is a complex number and b is a real number, then
- (A) the straight line $\frac{z-c}{a} + \frac{i(\bar{z}-\bar{c})}{a} = 0$ makes an angle of 45° with L and passes through a point c (where c is a complex number)
- (B) the straight line $\frac{z-c}{a} - \frac{i(\bar{z}-\bar{c})}{a} = 0$ makes an angle of 45° with L and passes through a point c (where c is a complex number)
- (C) the complex slope of the line L is $-\frac{a}{\bar{a}}$
- (D) the complex slope of the line L is $\frac{a}{\bar{a}}$
10. The function $f(x) = x^{1/3}(x-1)$
- (A) has two inflection points
- (B) has one point of extremum
- (C) is non – differentiable
- (D) Range of $f(x)$ is $[-3 \times 2^{-8/3}, \infty)$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Question Stem

Consider the locus of the complex number z in the Argand plane given by $\operatorname{Re}(z) - 2 = |z - 7 + 2i|$. Let $P(z_1)$, $Q(z_2)$. Be two complex numbers satisfying the given locus and also satisfying

$$\arg\left(\frac{z_1 - (2 + \alpha i)}{z_2 - (2 + \alpha i)}\right) = \frac{\pi}{2} (\alpha \in \mathbb{R}).$$

1. The minimum value of $PR \cdot QR$ where R represents the point $(7, -2)$ is
2. $\arg\left(\frac{z_1 - (7 - 2i)}{z_1 - (7 - 2i)}\right)$ equals

Paragraph for Questions 3 and 4

Question Stem

T is the region of the plane $x + y + z = 1$ with $x, y, z > 0$. S is the set of points (a, b, c) in T such that just two of the following three inequalities hold : $a \leq \frac{1}{2}, b \leq \frac{1}{3}, c \leq \frac{1}{6}$.

3. Area of the region T is
4. Area of the region S is

Paragraph for Questions 5 and 6

Question Stem

Consider the line $\frac{x-1}{2} = \frac{y}{-1} = \frac{z+1}{2}$ and the point $C(-1, 1, 2)$. Let D be the image of C in the line.

5. The distance C from the line is
6. The distance of the origin of the plane through the line and the point C is

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A, B, C and D cut a pack of 52 cards successively in the order given. If the person who cuts a spade first receives Rs. 350 and if the expectation of A is λ then $\left[\frac{\lambda}{64} \right]$ is equal to (where $[.]$ denotes greatest integer function) _____
8. Let $f(x) = \lim_{x \rightarrow \infty} \frac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$. If $f(x)$ is continuous for all $x \in \mathbb{R}$, then the value of $a + 8b$ is _____
9. A and B are two independent events. The probability that both A and B occur is $1/6$ and the probability that at least one of them occurs is $2/3$. Find $8P(A) + 9P(B)$ if $P(A) > P(B)$.

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

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